README for Using E-COSM 2021 Benchmark Package

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Thank you for downloading this package of open simulator package for *Real-time Energy Optimization of HEVs under Connected Environment: ECOSM 2021 Benchmark Problem and a Case Study* and we are looking forward to your solution for this challenging and meeting you in Tokyo, Japan in August 2021. Following information will show you how to use this package step by step.

1 Copyright

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2 Installation and File List

2.1 Installation

This software can only run in **MATLAB 2018a** so please install MATLAB 2018a in your PC before using this package. Following steps help you to install this package.

1. Unzip this package using decompression software firstly, such as WINRAR and WinZip.
2. Copy the unzipped folder in *Step 1* to the targeted folder, such as *C:/ECOSM2021* in the author's PC.
3. Open MATLAB 2018a, in Set Path of ENVIRONMENT, please *Add with Subfolders* to the folder *ECOSM2021*. Or you can also use the *Browse for folder* and find the *ECOSM2021* to open this package.

2.2 File List

In the Current Folder of MATLAB, you can find there are following files in the folder *Sophia_benchmark_model*:

- *Benchmark_model.slx*: the main file that containing the models of traffic scenario, HEV powertrain and a sample controller.
- *Benchmark_model.slxc*: automatically generated file, please do not delete it.
- *Initial_parameters.m*: the initial parameters for running the *Benchmark_model.slx*. Please run it before simulation. It is noted that in **line 49**, the variable of simulation time t_{end} =? should be defined by the challenger before simulation; otherwise, the simulation can not run successfully.
- *THSmodel_v4.slxp*: the encrypted file for Plant Model in Benchmark_model.slx.
- *DATA*: a folder with data of route slope, HEV parameters and traffic data collections.

Moreover, there are files in the folder *DATA*:

- *HEV_powertrain_parameters.mat*: the parameters of basic HEV powertrain, fuel consumption of engine, mechanical loss of motors and so on. The physical parameters and other map parameters listed in Table. 1 and Table. 2.
- *Route_intersection_destination_parameters.mat*: the position [m] of 26 intersections and destination from the departure.
- *Route_slope_parameters.mat*: the parameters of slope along the whole route, containing the information of distance [m], height [m] and slope [rad], which are represented as *Route_Distance*, *Route_Height* and *Route_Slope* in the Workspace, respectively.
- *Traffic_Data*: there are 10 *ECOSM2021_benchmark_problem_dataX.mat* files, which are generated through commercial traffic scenario simulation software. The detail meanings of each variable in the Workspace are listed in Table. 3.

Table 1: The meanings of HEV physical parameters

Parameter	Meaning	Unit	Parameter	Meaning	Unit	Parameter	Meaning	Unit
I_e	Engine inertia	kgm^2	R_s	Sun gear radius	mm	R_{f1}	Final gear 1 radius	mm
I_m	Motor inertia	kgm^2	R_r	Ring gear radius	mm	R_{f2}	Final gear 2 radius	mm
I_g	Generator inertia	kgm^2	R_m	Reduction gear radius	mm	R_t	Tire radius	mm
I_t	Tire inertia	kgm^2	R_{c1}	Counter gear 1 radius	mm	M	Vehicle mass	kg
Q_{batt}	Battery capacity	Ah	R_{c2}	Counter gear 2 radius	mm	$PbattMax$	Max battery power	kW

Table 2: Meanings of variables in HEV map parameters

Category	Variable	Description	Unit	Value
Engine	FuelConsmp_EngSpd	Engine speed vector in fuel consumption	rpm	14×1
	FuelConsmp_EngTrq	Engine torque vector in fuel consumption	Nm	1×30
	FuelConsmp_Rate_Matrix	Engine consumption rate	g/s	14×30
	Maxtorque_EngSpd	Engine speed vector in maximum engine torque	rpm	7×1
	Maxtorque_EngTrq	Engine torque vector in maximum engine torque	Nm	7×1
	MechanicalFriction_EngSpd	Engine speed vector in mechanical friction torque	rpm	34×1
	MechanicalFriction_EngTrq	Engine torque vector in mechanical friction torque	Nm	34×1
MG1	MG1_MaxMintorque_MaxTrq	Maximum torque vector	Nm	45×1
	MG1_MaxMintorque_MinTrq	Minimum torque vector	Nm	45×1
	MG1_MaxMintorque_Spd	Speed vector in max/min torque	rpm	45×1
	MG1_MechanicalLoss_Spd	Speed vector in mechanical loss	rpm	1×23
	MG1_MechanicalLoss_Trq	Torque vector in mechanical loss	Nm	23×1
	MG1_MechanicalLoss_Marix	Mechanical Loss	kW	23×23
MG2	MG2_MaxMintorque_MaxTrq	Maximum torque vector	Nm	72×1
	MG2_MaxMintorque_MinTrq	Minimum torque vector	Nm	72×1
	MG2_MaxMintorque_Spd	Speed vector in max/min torque	rpm	72×1
	MG2_MechanicalLoss_Spd	Speed vector in mechanical loss	rpm	1×35
	MG2_MechanicalLoss_Trq	Torque vector in mechanical loss	Nm	37×1
	MG2_MechanicalLoss_Matrix	Mechanical Loss	kW	37×35

3 Configuration and Operating

The overview capture of the simulator when you open *Benchmark_model.slx* is shown in Fig. 1. It consists of six parts, which are Traffic Scenario, Controller, Plant Model, Index Performance, Total Fuel Calculation and Index Display. The detail meanings of output signals for each block are also shown in the simulator.

3.1 Configuration

Before simulation, please check the model configuration parameters through  , which are set the same as Fig.2-Fig.11. It is noted that different configuration parameters may lead the nonsuccess of simulation.

3.2 Operating

In the *Benchmark_model.slx*, there is a block of Controller(Sample), the task of challengers is to design a controller to replace this sample one.

- Input of controller
  Any output signal of Plant Model is available to the controller and the detail meaning of each output signal is shown in *Benchmark_model.slx*. On the other hand, there are also V2V and V2I information from the block of Traffic Scenario that are available for the controller, including ego vehicle, sensor, V2V and V2I. The detail information can be seen in block of Traffic Scenario, which is open to the challengers.

Table 3: Meanings of variables in the Workspace

Category	Variable	Description	Unit	Note
Ego	EgoVhcl_Acceleration	Acceleration of ego vehicle	m/s^2	
	EgoVhcl_Distance	Distance of ego vehicle	m	
	EgoVhcl_Speed	Speed of ego vehicle	m/s	
Sensor	Sensor_FOL_Targ_Dtct	Detection flag	–	1 means detectable; 0 means undetectable
	Sensor_FOL_dDist	Distance headway	m	when detection flag is 0, headway is 0
V2V	PreVhcl_Acceleration	Acceleration of preceding vehicle	m/s^2	
	PreVhcl_Distance	Distance of preceding vehicle	m	
	PreVhcl_Speed	Speed of preceding vehicle	m/s	
V2I	TrfLight_State_Next1st	Traffic light phase of next 1st intersection	–	0 no signal; 1 green light; 3 red light
	TrfLight_tRemain_Next1st	Traffic light timing of next 1st intersection	s	no next 1st traffic light, value is 0
	TrfLight_State_Next2nd	Traffic light phase of next 2nd intersection	–	0 no signal; 1 green light; 3 red light
	TrfLight_tRemain_Next2nd	Traffic light timing of next 2nd intersection	s	no next 2nd traffic light, value is 0
	TrfLight_State_TL(1-26)	Traffic light phase of (1-26) intersection	–	0 off light; 1 green light; 3 red light
	TrfLight_tRemain_TL(1-26)	Traffic light timing of (1-26) intersection	s	0 off light, free pass
GPS	Route_Slope	Slope of the whole route	rad	the collected data is dependent on time
System	System_Time	The simulation time in traffic simulator	s	
	Traffic_data_SamplingTime	The sampling time for data collection	s	

- Output of controller
  There are four variables should be controlled, which are signals of engine torque, motor torque, generator torque and mechanical braking force for braking the vehicle.

4 Bugs and User FAQ

To deal with the inevitable bugs in this package, the author will update the bugs information and give the solution through this Readme. Moreover, the FAQ from the challenger will also be updated in this Readme. For more information, please find it through $http://www.shenlab.jp/ecosm2021/index.html$.

4.1 Bugs

4.1.1 20200715

None

4.2 User FAQ

4.2.1 20200715

None

5 Acknowledgements

The authors would like to thank Toyota Motor Corporation, Japan, for technical support.

6 Contact

If there is any problem of this open software or any question about the benchmark problem, please do not hesitate to contact Dr. Fuguo XU by *fuguoxu@sophia.ac.jp*.

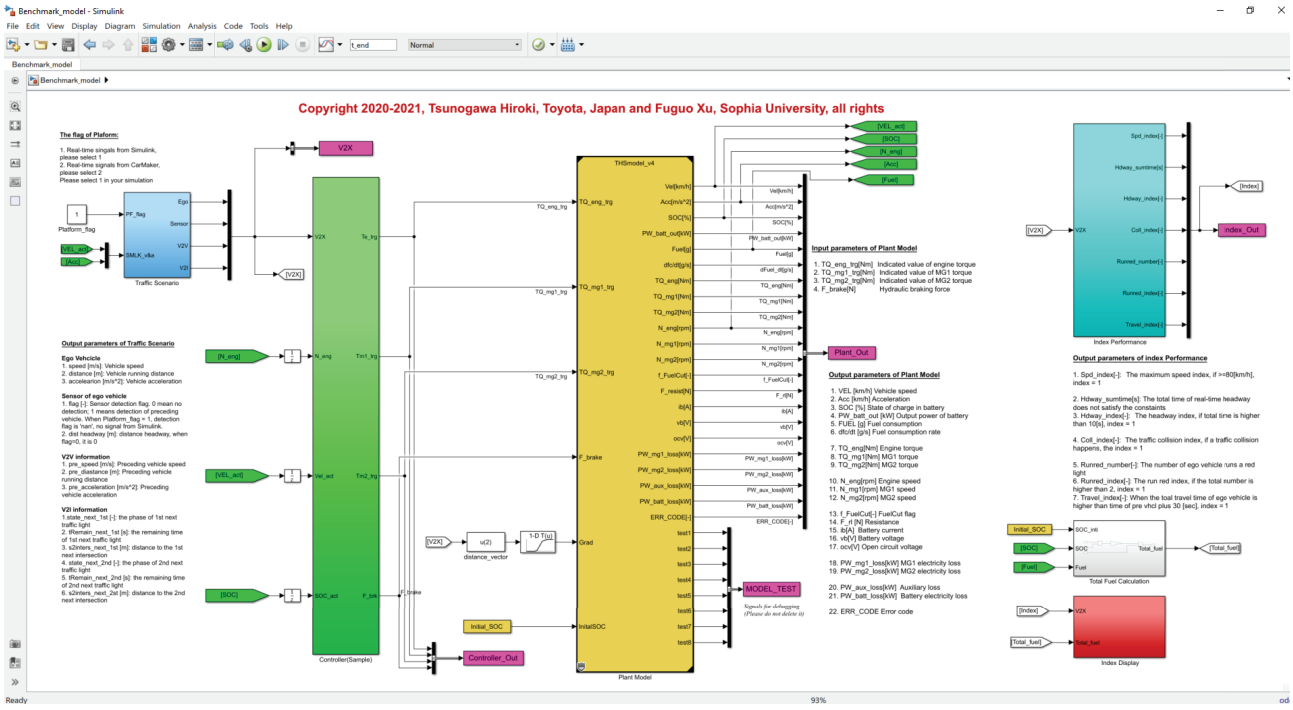


Figure 1: The overview capture of the simulator in MATLAB/Simulink

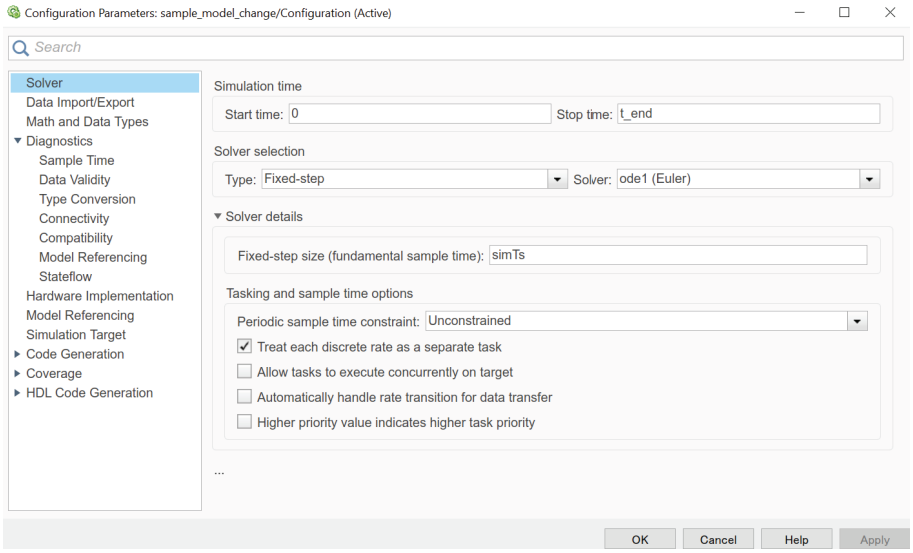


Figure 2: Model configuration parameters 1

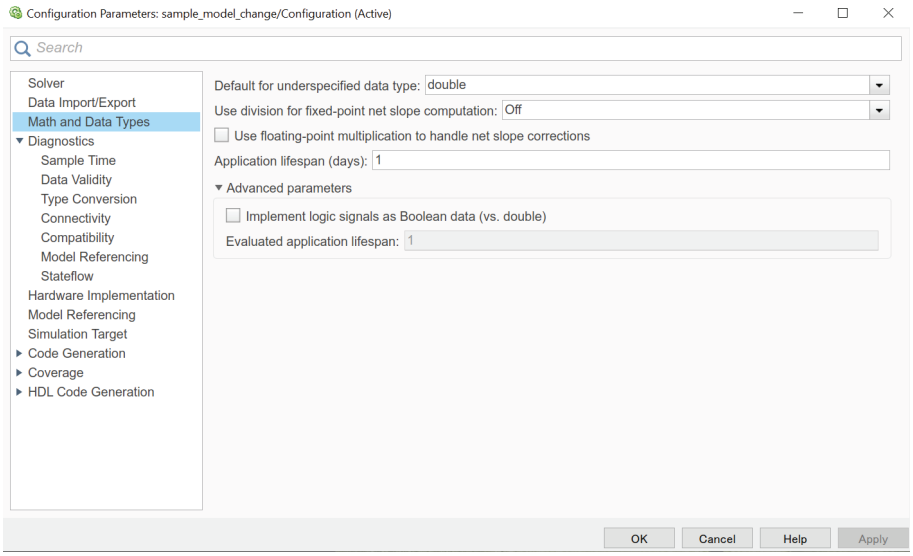


Figure 3: Model configuration parameters 2

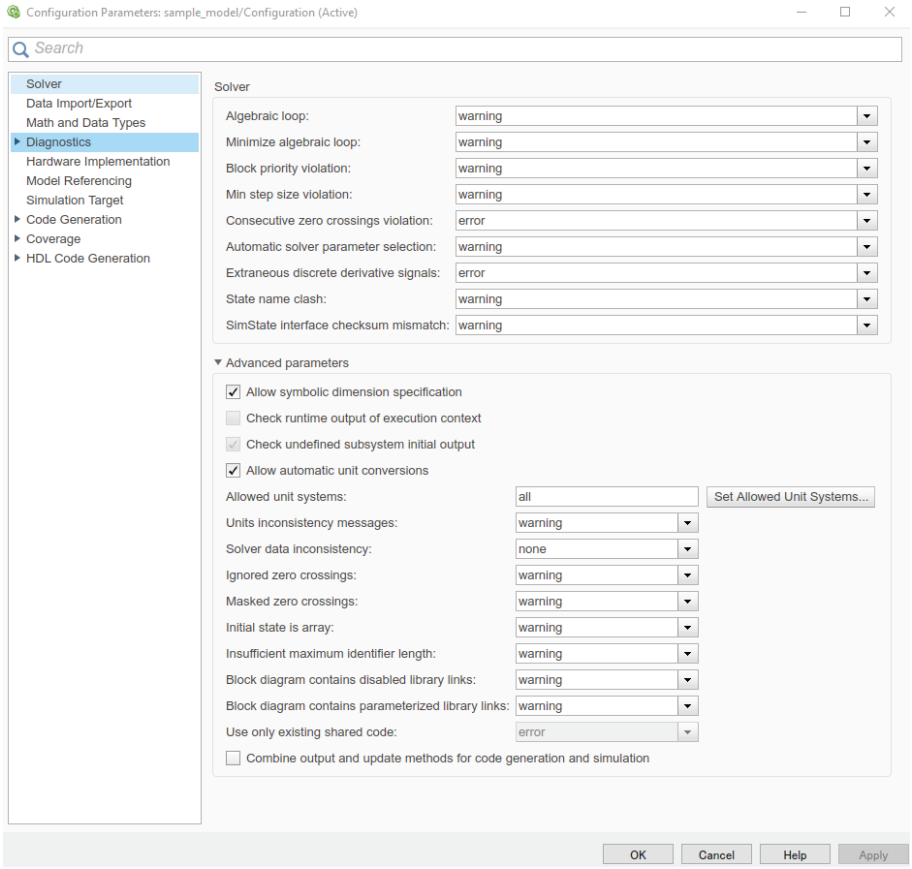


Figure 4: Model configuration parameters 3

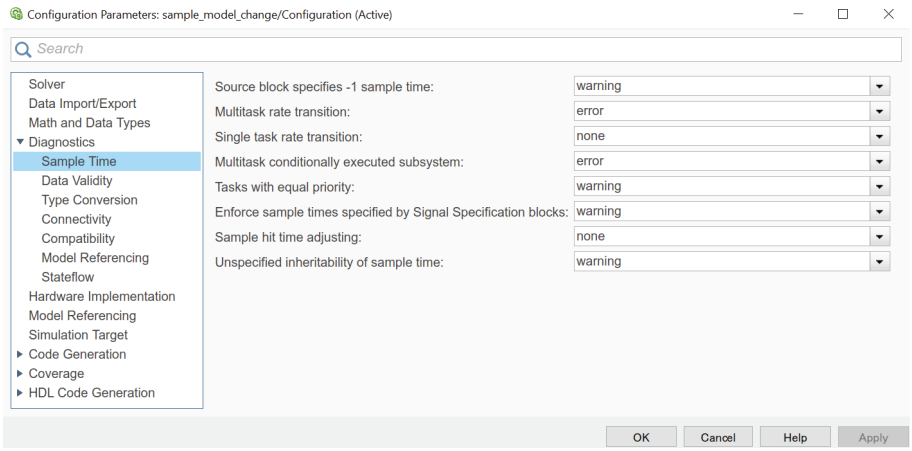


Figure 5: Model configuration parameters 4

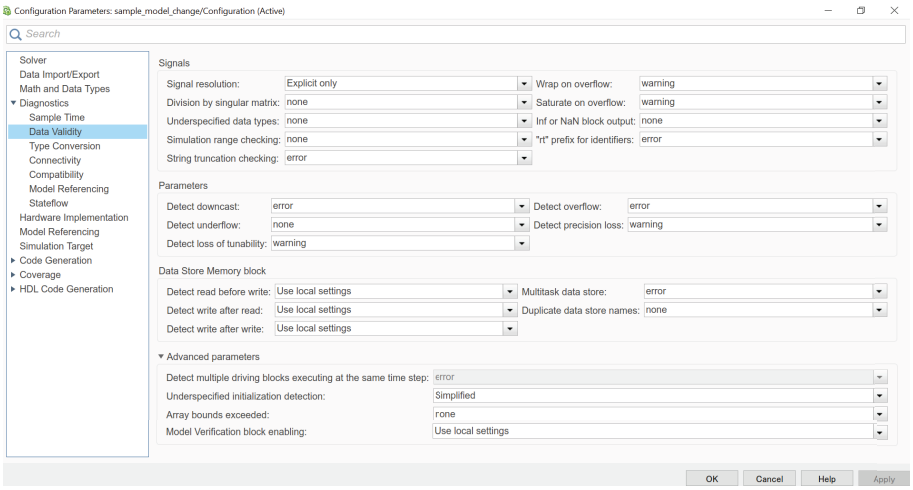


Figure 6: Model configuration parameters 5

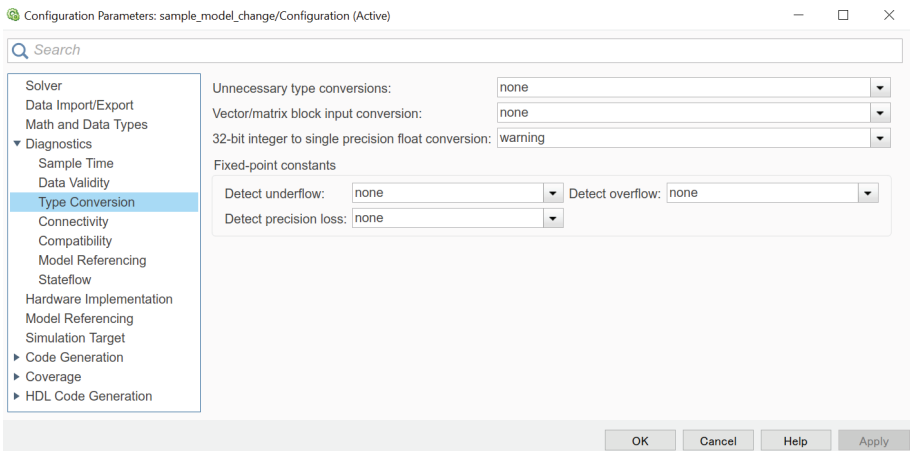


Figure 7: Model configuration parameters 6

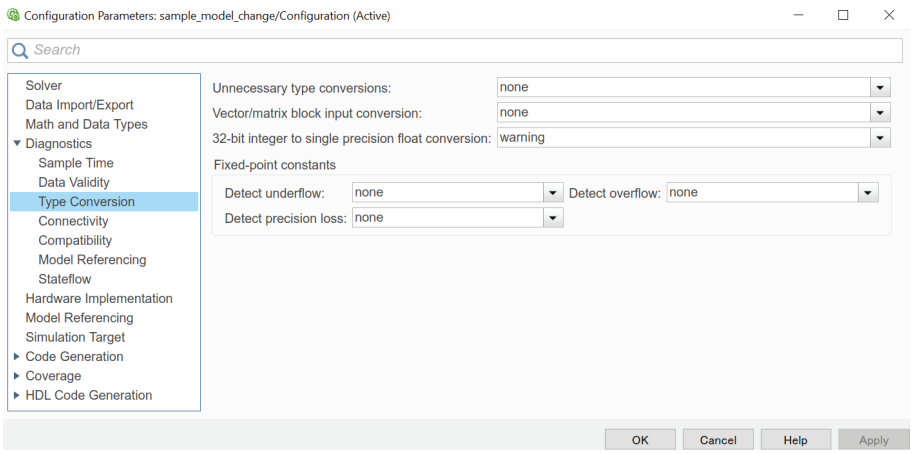


Figure 8: Model configuration parameters 7

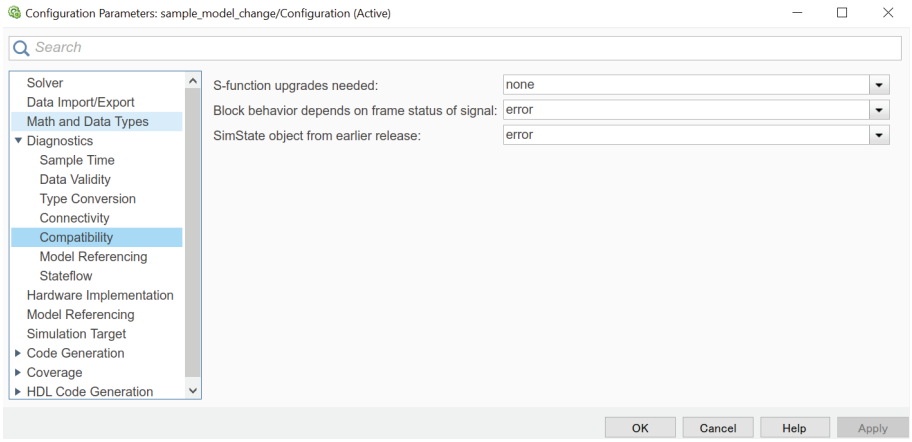


Figure 9: Model configuration parameters 8

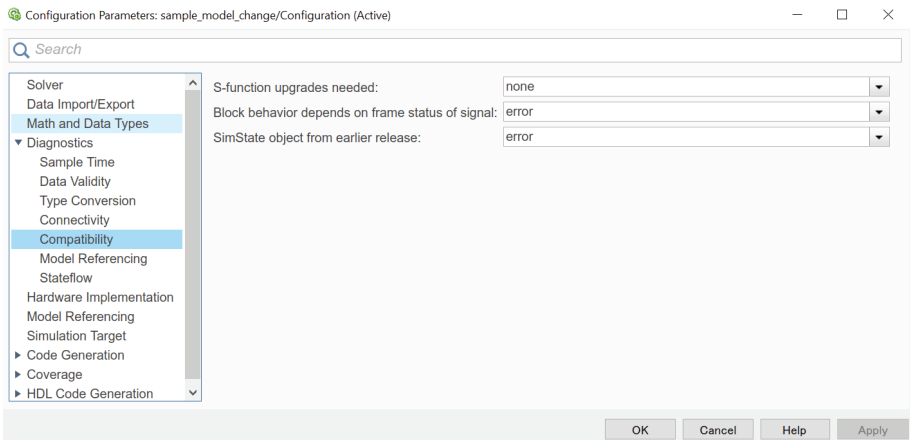


Figure 10: Model configuration parameters 9

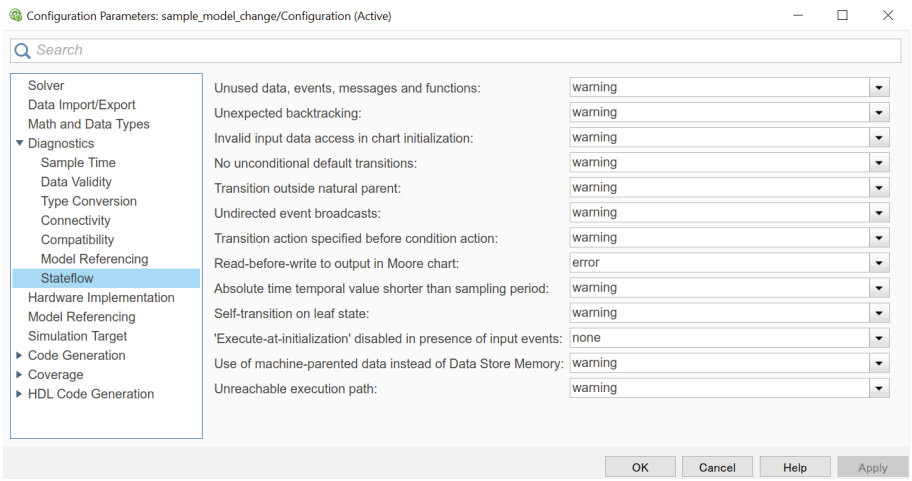


Figure 11: Model configuration parameters 10